

C: SquareDiff

Source file name: `sqdf.c`, `sqdf.cpp`, `sqdf.java`, or `sqdf.py`

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Taking a break from her advanced number theory research, Alana began playing with square differences:

- $1^2 - 0^2 = 1$
- $6^2 - 4^2 = 20$
- $10^2 - 5^2 = 75$

However, she has not found two integers x and y such that $x^2 - y^2 = 2$. It might be the case that such a pair does not exist. This made her wonder if there are other numbers that are equally rare.

Alana wants to determine whether a given integer can be expressed as the difference of two squares of integers. Can you help her by writing a program to solve this problem?

Input

The input consists of several test cases. Each test case is defined by a single line containing a positive integer N ($0 < N < 40\,000$), which is the number to test. The input ends with a line containing 0, which should not be processed.

The input must be read from standard input.

Output

For each test case, print a single line with one character: 'Y' if N can be expressed as the difference of two squares of integers and 'N' otherwise.

The output must be written to standard output.

Sample Input	Sample Output
1	Y
2	N
3	Y
20	Y
0	